

TESTING THE FIT OF A DISTRIBUTION.

WE GENERATE MANY SAMPLES OF A R.V.
AND LOOK AT THE HISTOGRAM.

HOW WELL DOES THE SAMPLE HISTOGRAM
MATCH THE EXPECTED PDF / PMF?

~~FIX~~

EX LEGO COLORS

COLOR	R	W	Y	B	BLACK K	TOTAL
OBSERVED	48	37	48	25	42	= 200

HYPOTHESIS ABOUT TRUE PMF:

COLOR	R	W	Y	B	K
PMF	0.3	0.2	0.2	0.15	0.15
EXPECTED	60	40	40	30	30

WE SET A SIGNIFICANCE LEVEL α AND

THEN, CAN WE ACCEPT/REJECT HYPOTHESIS AT THE $\alpha\%$ LEVEL?

WE CREATE THE TEST STATISTIC:

$$D^2 = \sum_{k=1}^{K = \# \text{ OUTCOMES}} \frac{1}{E_k} (O_k - E_k)^2$$

↑ ↑
OBSERVED EXPECTED
OUTCOME = k OUTCOME = k

IN EXAMPLE,

$$D^2 = \frac{(60 - 48)^2}{60} + \frac{(37 - 40)^2}{40} + \dots = 9.86.$$

WE COMPARE THIS VALUE TO A χ^2 ("CHI-SQUARED")
RANDOM VARIABLE WITH $K-1$
DEGREES OF FREEDOM. (LOOK IN A TABLE)

* [IF THE D^2 VALUE IS ABOVE THE
NUMBER IN THE TABLE, WE CAN REJECT
THE HYPOTHESIS AT THE $\alpha\%$ LEVEL.
OTHERWISE THE DATA IS CONSISTENT AT THIS
LEVEL.

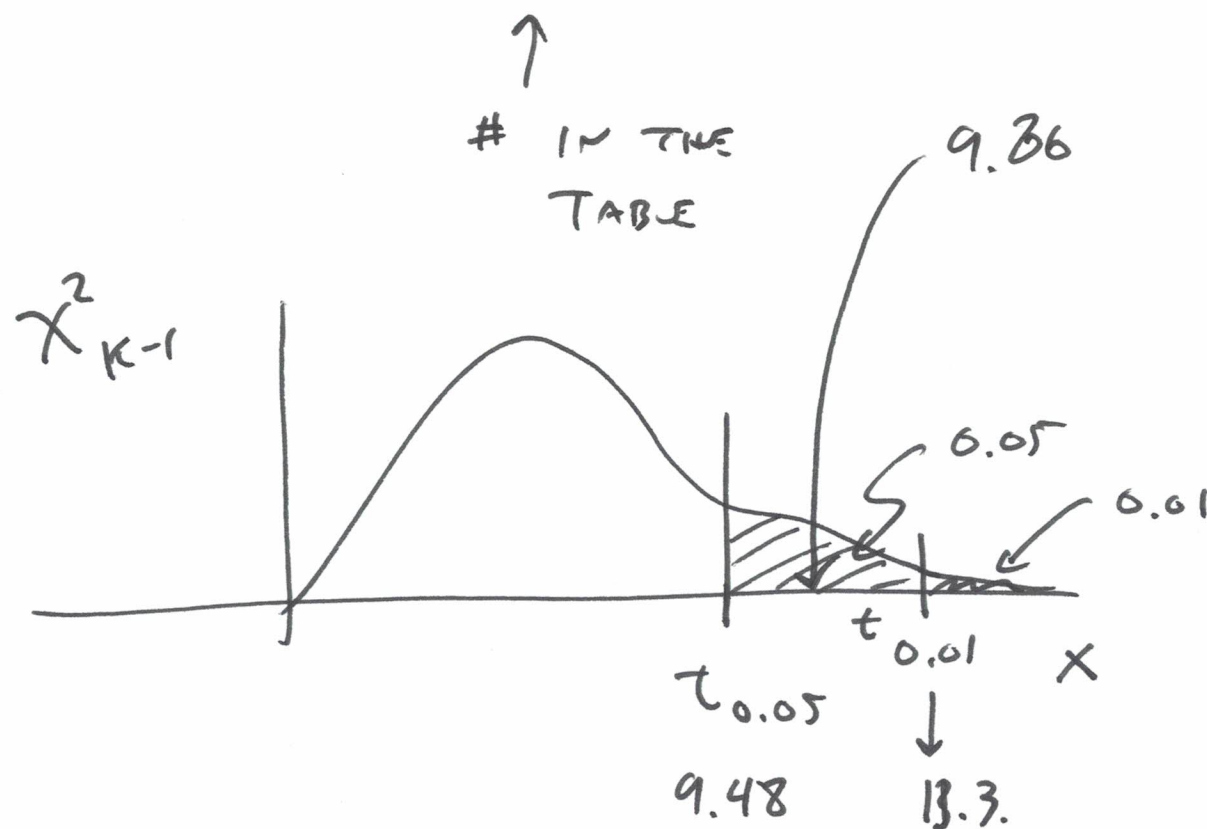
IN EXAMPLE, $D^2 = 9.86$.

SINCE $D^2 = 9.86 < 13.277$, WE CAN
ACCEPT THE HYPOTHESIS AT THE $\alpha = 1\%$
LEVEL.

BUT $D^2 = 9.86 > 9.488$, WE WOULD REJECT
THE HYPOTHESIS AT THE 5% LEVEL.

α IS LIKE A THRESHOLD:

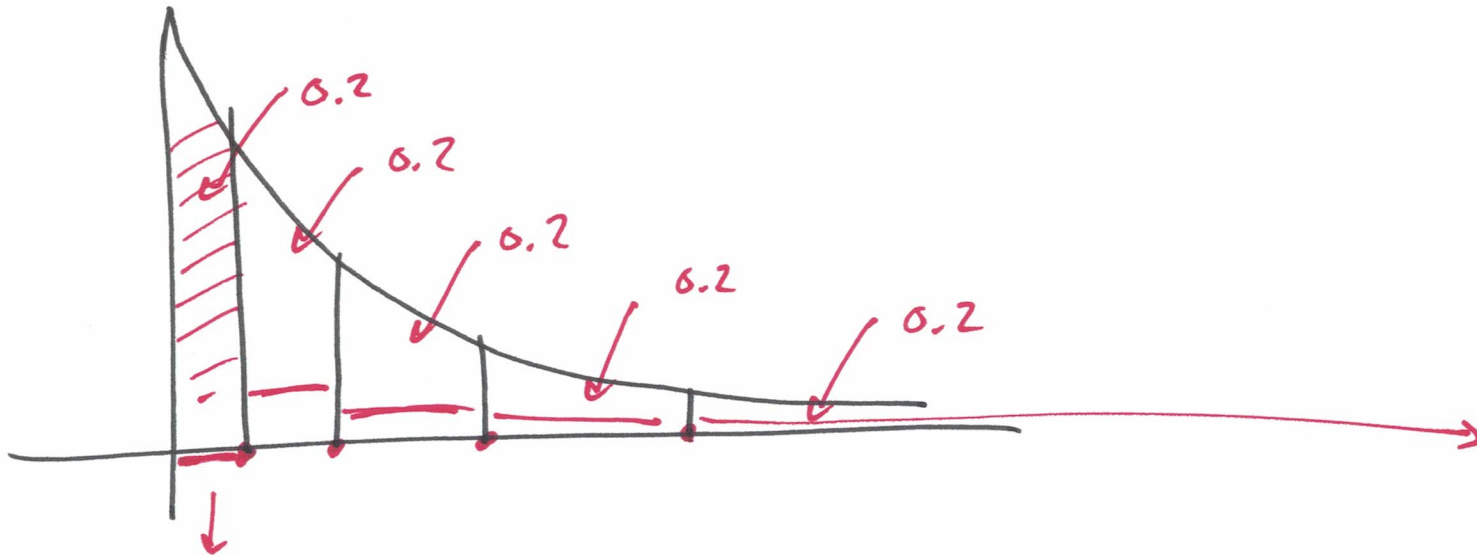
$$P(D^2 > t_\alpha) = \alpha.$$



CONSISTENT AT
~~9%~~ 1%
NOT CONSISTENT
(REJECT)
AT 5%.

KEY TAKEAWAY: THIS D^2 VALUE SHOULD BE
LOW FOR A GOOD FIT.

TO TEST THE FIT OF A
CONTINUOUS R.V., CREATE EQUALLY
PROBABLE BINS FOR THE OUTCOMES:



SIGNIFICANCE TESTING: COMPARING AVG. SAMPLE TO
A PDF.

D^2 TESTING: COMPARING ENTIRE SAMPLE PDF
TO TARGET PDF.

RANDOM NUMBER GENERATION:

ANY PROGRAMMING LANGUAGE PROVIDES:

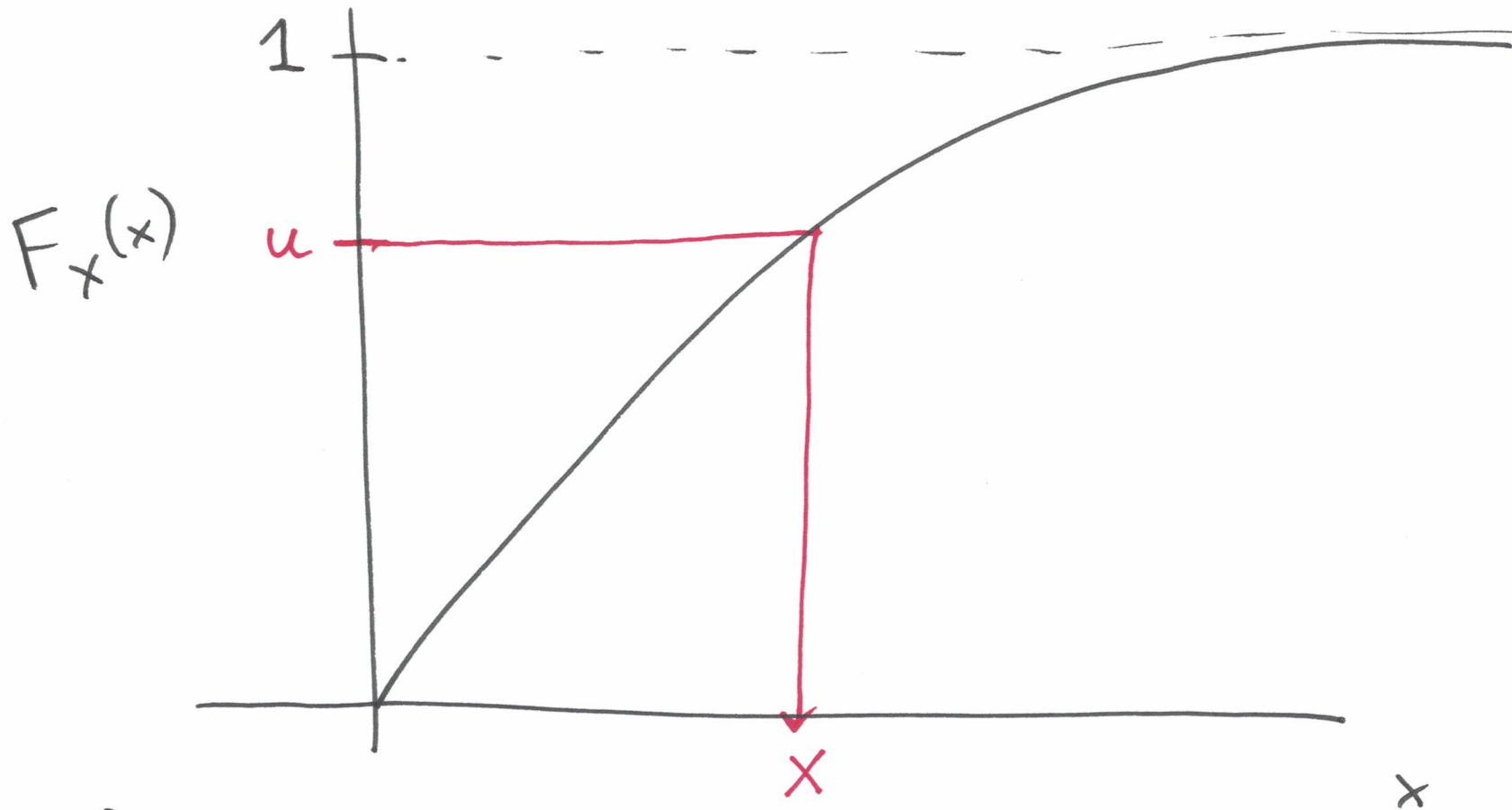
$\text{rand} \rightarrow \text{UNIFORM } [0, 1] \text{ SAMPLE} \rightarrow U$

(IN PRACTICE THIS ~~IS~~ IS A
DETERMINISTIC PROCESS STARTING FROM
A "SEED", INDISTINGUISHABLE FROM
~~BEING~~ BEING TOTALLY RANDOM)

How TO CONVERT THIS INTO SAMPLING

ANY RANDOM VARIABLE? $\rightarrow X$

IT'S REALLY EASY IF WE KNOW
THE CDF OF X :



- 1) GENERATE U ON $[0, 1]$
- 2) LET X BE $F_X^{-1}(u)$

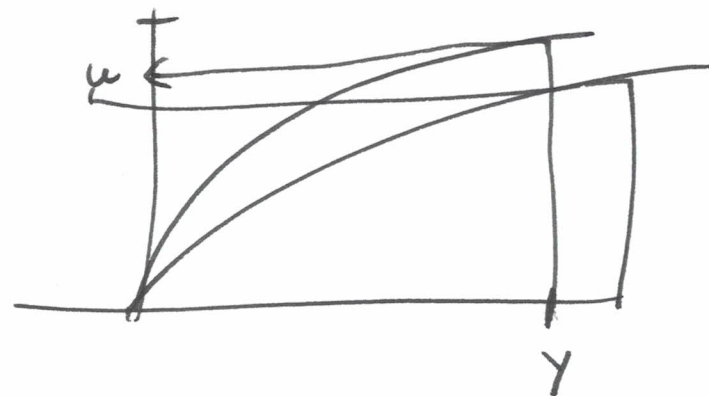
EX EXPONENTIAL WITH $\lambda = 1$.

$$F_X(x) = 1 - e^{-x}$$

$$u = 1 - e^{-x}$$

$$u - 1 = -e^{-x}$$

$$x = -\ln(1 - u)$$



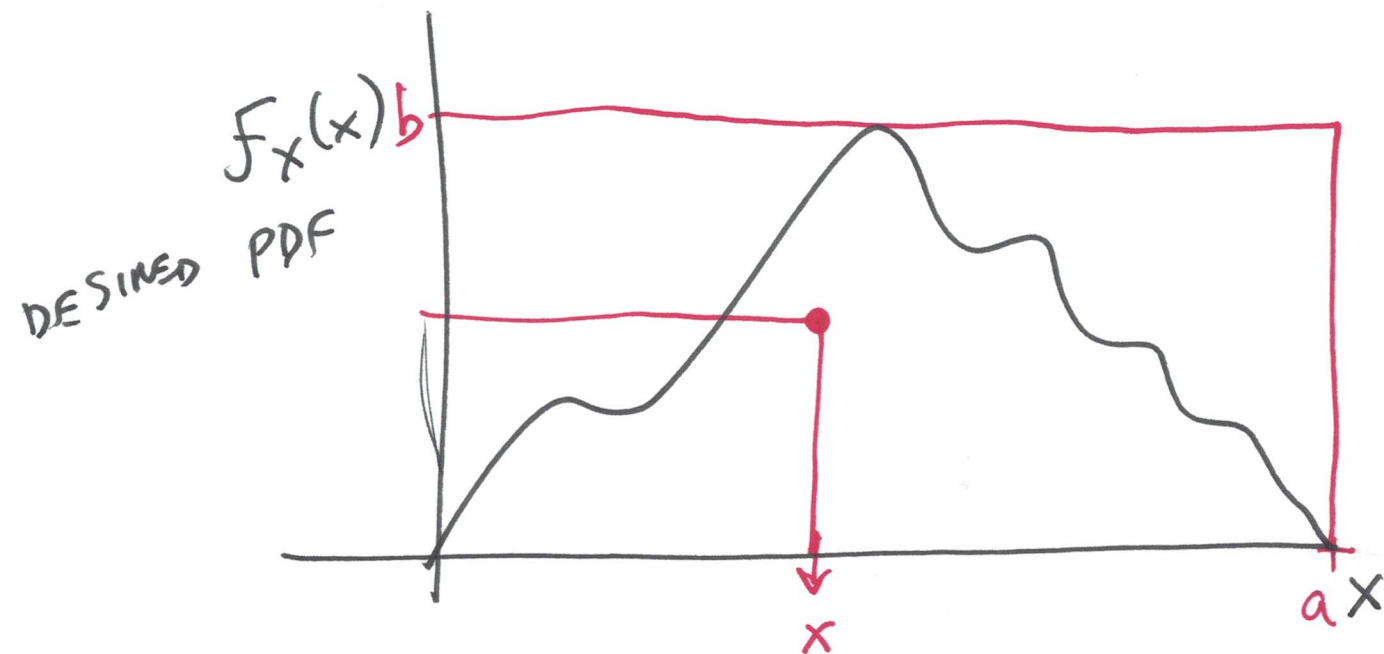
IF I CAN GENERATE Y THAT ISN'T
UNIFORM ON $[0, 1]$

$$Y \rightarrow U \rightarrow X$$

$$u = F_Y(y) \quad x = F_X^{-1}(u)$$

IT'S NOT ALWAYS POSSIBLE TO WRITE
DOWN $F_X^{-1}(x)$. (CDF MAY NOT
BE EASILY INVERTIBLE.)

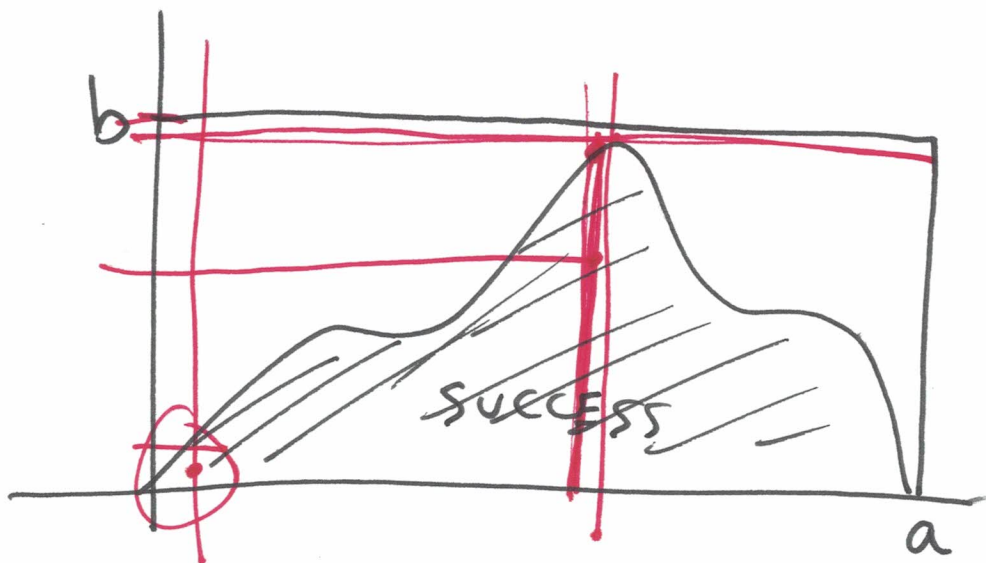
IN THIS CASE WE CAN USE
THE REJECTION METHOD: (IF PDF HAS
BOUNDED
OUTCOMES



IF DART THROWN
IS UNDER THE
PDF CURVE,
ACCEPT x .
OTHERWISE TRY
AGAIN.

- 1) GENERATE U ON $[0, a]$ UNIFORM
 BIGGEST VALUE OF X
- 2) GENERATE Y ON $[0, b]$ UNIFORM
 BIGGEST VALUE OF $f_X(x)$.
- 3) IF $Y \leq f_X(u)$ ACCEPT U AS
A SAMPLE OF X . OTHERWISE TRY AGAIN.

WHY DOES THIS WORK?



$$P(\text{SUCCESS}) = \frac{1}{ab}$$

SO SUCCEEDING IS A GEOMETRIC P.V.

$$\text{WITH } p = \frac{1}{ab} \quad E(\text{TRIALS}) = \frac{1}{p} = ab$$

$$f_x(x | \text{ACCEPT}) = \frac{P(\text{ACCEPT} | x) f_x(x)}{P(\text{ACCEPT})}$$

$\uparrow$
 PDF OF
 ACCEPTED VALUES

$\downarrow$
 $\frac{1}{ab}$

$$f_x(x) = \frac{1}{a} \quad \text{SINCE } x \text{ IS UNIFORM ON } [0, a].$$

~~Find~~ $P(\text{ACCEPT} | x) = \frac{f_x(x)}{b}$

$$\frac{f_x(x)}{b} \cdot \frac{1}{a} \div \frac{1}{ab} = f_x(x).$$

THERE'S ALSO PROBABLY A

randn $\rightarrow$ GAUSSIAN $\mu=0$
 $\sigma=1$.

WE NEED EFFICIENT / PREDICTABLE WAYS
OF GENERATING THESE.

FOR EXAMPLE, THERE'S A WAY TO
CONVERT 2 UNIFORM R.V.S. INTO
2 GAUSSIAN R.V.S.

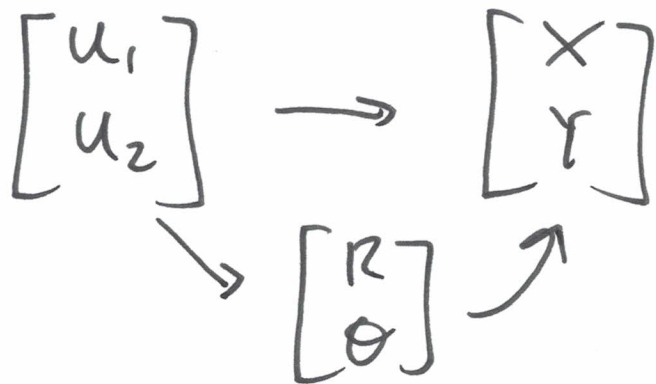
1) GENERATE U_1, U_2 UNIFORM IN $[0, 1]$

2) SET $R = \sqrt{-2 \log U_1}$ $\Theta = 2\pi U_2$

3) $X = R \cos \Theta$, $Y = R \sin \Theta$

X AND Y ARE INDEPENDENT $\mu=0, \sigma=1$
GAUSSIAN RANDOM VARIABLES.

HOW COULD WE SHOW THIS IS TRUE?



IN GENERAL, $(X, Y) \rightarrow (w, z)$

$$X = h_1(w, z)$$

$$Y = h_2(w, z)$$

$$f_{wz}(w, z) = f_{xy}(h_1(w, z), h_2(w, z)) \cdot J_h(w, z)$$

$\begin{matrix} X & Y \\ \downarrow & \downarrow \end{matrix}$

$$J_h = \left| \begin{bmatrix} \frac{\partial h_1}{\partial w} & \frac{\partial h_1}{\partial z} \\ \frac{\partial h_2}{\partial w} & \frac{\partial h_2}{\partial z} \end{bmatrix} \right| \leftarrow \text{THE JACOBIAN.}$$